

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-61040707})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}2713 + \mathbf{Z}(751 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 15260177 = 0$, of discriminant -61040707 . Class group invariants $(20 \ 10 \ 5)$, class number 1000; the three stored generator ideals (HNF) are $\begin{pmatrix} 193 & 45 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 761 & 748 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 596)y^3 + (-15s - 91547)y^2 + (-15s + 493029)y + (-594s + 4156434)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 1194y^8 - 184302y^7 + 16785141y^6 - 559703590y^5 \\ & + 11937823312y^4 - 96580412844y^3 - 242586001017y^2 + 43700661455 \\ & 76y + 22657814486532 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right)s, \quad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3223534230000 = 2^4 \cdot 3^2 \cdot 5^4 \cdot 7 \cdot 5116721$:

$$\begin{pmatrix} 2149022820 & 1785426780 & 1137977520 & 1142896165 & 1067502673 & 857198691 & 1784352178 & 1987745796 & 384106826 & 1259346964 \\ 0 & 15 & 10 & 0 & 3 & 11 & 0 & 1 & 5 & 11 \\ 0 & 0 & 20 & 15 & 11 & 4 & 1 & 0 & 12 & 0 \\ 0 & 0 & 0 & 5 & 1 & 4 & 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 269 factors with signed exponents of absolute value up to 82539193764; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 596)y^3 + (-15s - 91547)y^2 + (-15s + 493029)y + (-594s + 4156434)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 1194y^8 - 184302y^7 + 16785141y^6 - 559703590y^5 \\ + 11937823312y^4 - 96580412844y^3 - 242586001017y^2 + 43700661455 \\ 76y + 22657814486532$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right)s, \quad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1283346364716 = 2^2 \cdot 3^2 \cdot 13 \cdot 2742193087$:

$$\begin{pmatrix} 213891060786 & 35917654728 & 87186283752 & 3019265520 & 82720395321 & 164964566711 & 31410345170 & 16072540820 & 180702219631 & 127163484285 \\ 0 & 3 & 2 & 2 & 0 & 2 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 233 factors with signed exponents of absolute value up to 242462; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 4 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 596)y^3 + (-15s - 91547)y^2 + (-15s + 493029)y + (-594s + 4156434)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 1194y^8 - 184302y^7 + 16785141y^6 - 559703590y^5 \\ + 11937823312y^4 - 96580412844y^3 - 242586001017y^2 + 43700661455 \\ 76y + 22657814486532$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + \left(\frac{10043}{2626056128537293}\right) s, \quad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $27418800636688 = 2^4 \cdot 19 \cdot 229 \cdot 7699 \cdot 51157$:

$$\begin{pmatrix} 3427350079586 & 2837092662522 & 2608895173368 & 2725223177634 & 273929779878 & 1666951123989 & 103912233536 & 721957175337 & 1303710000025 & 1625182099598 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 274 factors with signed exponents of absolute value up to 84882662179; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 22y^3 + (22s - 11)y^2 + 2174656y + (-4480s + 2240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 44y^8 + 4349796y^6 + 7481610411y^4 + 1721042677376y^2 + 306277851443200$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right) s, \quad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $37447973017268 = 2^2 \cdot 11^2 \cdot 193 \cdot 400890389$:

$$\begin{pmatrix} 1702180591694 & 258067167974 & 682605261550 & 694633268330 & 1349242601298 & 1074882142914 & 601175944021 & 671773640265 & 690264828372 & 735697201907 \\ 0 & 22 & 0 & 12 & 2 & 11 & 21 & 1 & 9 & 18 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 203 factors with signed exponents of absolute value up to 2149542977247342950178242574753; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 22y^3 + (22s - 11)y^2 + 2174656y + (-4480s + 2240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 44y^8 + 4349796y^6 + 7481610411y^4 + 1721042677376y^2 + 306277851443200$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right)s, \quad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $13872356136639 = 3^2 \cdot 7^2 \cdot 11 \cdot 13 \cdot 2557 \cdot 86029$:

$$\begin{pmatrix} 660588387459 & 361590177312 & 4905090085 & 285548101967 & 261624281266 & 77192817684 & 281291431946 & 623907613571 & 460260605917 & 238076830007 \\ 0 & 21 & 10 & 0 & 14 & 20 & 6 & 8 & 6 & 4 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 22y^3 + (22s - 11)y^2 + 2174656y + (-4480s + 2240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 44y^8 + 4349796y^6 + 7481610411y^4 + 1721042677376y^2 + 306277851443200$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + \left(\frac{10043}{2626056128537293}\right)s, \quad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18651079917568 = 2^{14} \cdot 7^2 \cdot 53 \cdot 438341$:

$$\begin{pmatrix} 5203984352 & 726604368 & 3963481616 & 1450420208 & 3784021470 & 5149115704 & 1033110481 & 4796696563 & 1621971298 & 2214930015 \\ 0 & 14 & 12 & 10 & 10 & 10 & 2 & 11 & 7 & 4 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 32 & 30 & 14 & 10 & 23 & 31 & 15 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 1149)y^3 + (-13s + 63831)y^2 + (908s - 547750)y + (10640s - 53110896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 2290y^8 + 134546y^7 + 15103988y^6 + 147104694y^5 \\ & - 19483077347y^4 - 632770881834y^3 + 1880550308684y^2 + 352989736 \\ & 250912y + 4547800708088576 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 1 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right)s, \quad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $36819507031524 = 2^2 \cdot 3^6 \cdot 7 \cdot 2713 \cdot 664879$:

$$\begin{pmatrix} 227280907602 & 151722996201 & 1762524141 & 99659990535 & 40823140242 & 41942221512 & 88198242124 & 141706878240 & 108060483270 & 14985868604 \\ 0 & 9 & 3 & 5 & 3 & 5 & 8 & 0 & 8 & 4 \\ 0 & 0 & 6 & 1 & 3 & 5 & 5 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 361 factors with signed exponents of absolute value up to 4515646560754155577761446960; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 1149)y^3 + (-13s + 63831)y^2 + (908s - 547750)y + (10640s - 53110896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 2290y^8 + 134546y^7 + 15103988y^6 + 147104694y^5 \\ & - 19483077347y^4 - 632770881834y^3 + 1880550308684y^2 + 352989736 \\ & 250912y + 4547800708088576 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 1 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right)s, \quad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7963940593728 = 2^6 \cdot 3^2 \cdot 7^4 \cdot 277 \cdot 20789$:

$$\begin{pmatrix} 483718452 & 227867976 & 132130880 & 211282414 & 410684344 & 49693198 & 120113279 & 422541736 & 120815308 & 141680722 \\ 0 & 84 & 70 & 42 & 21 & 0 & 28 & 70 & 19 & 32 \\ 0 & 0 & 14 & 0 & 8 & 8 & 13 & 6 & 2 & 3 \\ 0 & 0 & 0 & 14 & 7 & 8 & 1 & 0 & 1 & 12 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 359 factors with signed exponents of absolute value up to 3801377806980182705883337536; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 1149)y^3 + (-13s + 63831)y^2 + (908s - 547750)y + (10640s - 53110896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 2290y^8 + 134546y^7 + 15103988y^6 + 147104694y^5 \\ & - 19483077347y^4 - 632770881834y^3 + 1880550308684y^2 + 352989736 \\ & 250912y + 4547800708088576 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 1 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + \left(\frac{10043}{2626056128537293}\right)s, \quad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25661276390144 = 2^8 \cdot 107 \cdot 7309 \cdot 128173$:

$$\begin{pmatrix} 400957443596 & 13388486968 & 231719016696 & 331499651696 & 55411858338 & 308723671139 & 102041695281 & 1934600540 & 346085036866 & 292247556734 \\ 0 & 4 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 4 & 0 & 1 & 0 & 1 & 1 & 2 & 3 \\ 0 & 0 & 0 & 4 & 1 & 0 & 3 & 3 & 3 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 300 factors with signed exponents of absolute value up to 38367884847714121030263148024; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 849)y^3 + (15s + 52354)y^2 + (-159s - 1104054)y + (-2150s + 5958260)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 1698y^8 + 106422y^7 + 13668837y^6 - 532644854y^5 \\ & + 12892011104y^4 - 132921712908y^3 + 1244476198619y^2 - 272167220 \\ & 4420y + 106028220151100 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right)s, \quad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $42055285409564 = 2^2 \cdot 7 \cdot 1501974478913$:

$$\begin{pmatrix} 21027642704782 & 6687803274550 & 2206557951661 & 15759836842411 & 9116092259351 & 2531545033149 & 4391281621681 & 16923864274922 & 7455345516788 & 12073422506684 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 320 factors with signed exponents of absolute value up to 76540389117; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 2 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 849)y^3 + (15s + 52354)y^2 + (-159s - 1104054)y + (-2150s + 5958260)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 1698y^8 + 106422y^7 + 13668837y^6 - 532644854y^5 \\ + 12892011104y^4 - 132921712908y^3 + 1244476198619y^2 - 272167220 \\ 4420y + 106028220151100$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right)s, \quad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $61701674840064 = 2^{12} \cdot 3^2 \cdot 7 \cdot 31^2 \cdot 248813$:

$$\begin{pmatrix} 1295818104 & 366026052 & 1161697736 & 793935487 & 479271616 & 362044164 & 694620372 & 351302106 & 959487044 & 1245163039 \\ 0 & 372 & 8 & 136 & 250 & 289 & 312 & 124 & 41 & 117 \\ 0 & 0 & 8 & 4 & 2 & 0 & 4 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 320 factors with signed exponents of absolute value up to 7068151712; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 849)y^3 + (15s + 52354)y^2 + (-159s - 1104054)y + (-2150s + 5958260)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 1698y^8 + 106422y^7 + 13668837y^6 - 532644854y^5 \\ + 12892011104y^4 - 132921712908y^3 + 1244476198619y^2 - 272167220 \\ 4420y + 106028220151100$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + \left(\frac{10043}{2626056128537293}\right) s, \quad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $11640919262427 = 3^2 \cdot 7 \cdot 13 \cdot 61 \cdot 139 \cdot 953 \cdot 1759$:

$$\begin{pmatrix} 3880306420809 & 2770947584832 & 257541559789 & 646517670096 & 2228094089263 & 1694106045929 & 124414833767 & 539641311615 & 2306317086218 & 3135086743723 \\ 0 & 3 & 1 & 0 & 1 & 0 & 2 & 2 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 319 factors with signed exponents of absolute value up to 149665316424; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 532y^3 + (-16s + 80745)y^2 + (298s + 716171)y + (-15842s + 71652976)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 1065y^8 + 160410y^7 + 1554190y^6 + 227761638y^5 + \\ & 11043942787y^4 + 46376348712y^3 + 21173147731929y^2 - 4144245356 \\ & 8606y + 8962844655357412 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right) s, \quad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1413524858297 = 29 \cdot 48742236493$:

$$\begin{pmatrix} 1413524858297 & 694132558939 & 224372580134 & 570836977290 & 920159279216 & 1151675387126 & 78209000894 & 344919351288 & 211108289585 & 1005992030845 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 301 factors with signed exponents of absolute value up to 1440721437012; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 1 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 532y^3 + (-16s + 80745)y^2 + (298s + 716171)y + (-15842s + 71652976)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 1065y^8 + 160410y^7 + 1554190y^6 + 227761638y^5 + \\ & 11043942787y^4 + 46376348712y^3 + 21173147731929y^2 - 4144245356 \\ & 8606y + 8962844655357412 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right)s, \quad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $32646359966976 = 2^8 \cdot 3^2 \cdot 7^5 \cdot 843067$:

$$\begin{pmatrix} 991446792 & 232200906 & 648221154 & 384666779 & 652330094 & 133447866 & 763883919 & 614873529 & 920565661 & 928259404 \\ 0 & 294 & 238 & 130 & 260 & 42 & 68 & 247 & 282 & 17 \\ 0 & 0 & 56 & 31 & 20 & 12 & 40 & 46 & 10 & 50 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 241 factors with signed exponents of absolute value up to 166439366787; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 532y^3 + (-16s + 80745)y^2 + (298s + 716171)y + (-15842s + 71652976)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 1065y^8 + 160410y^7 + 1554190y^6 + 227761638y^5 + 11043942787y^4 + 46376348712y^3 + 21173147731929y^2 - 41442453568606y + 8962844655357412$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + \left(\frac{10043}{2626056128537293}\right)s, \quad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $19500028948944 = 2^4 \cdot 3^2 \cdot 135416867701$:

$$\begin{pmatrix} 812501206206 & 386452438650 & 442758507896 & 704577392328 & 3164248961 & 68346854312 & 340990993512 & 606640849321 & 548026715662 & 419588420064 \\ 0 & 6 & 2 & 4 & 3 & 2 & 1 & 3 & 4 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 231 factors with signed exponents of absolute value up to 271011566; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1832)y^3 + (22s - 19966)y^2 + (-93s + 666465)y + (-360s - 1976130)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 3661y^8 - 32580y^7 + 20030890y^6 - 604930996y^5 + \\ & 8188001885y^4 - 70810902186y^3 + 413252218233y^2 - 1612277656290 \\ & *y + 5883520122900 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right)s, \quad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $28278521334297 = 3^8 \cdot 13 \cdot 193 \cdot 1717853$:

$$\begin{pmatrix} 38790838593 & 6164566551 & 24904683843 & 21111862527 & 31847022519 & 32507928810 & 36947507045 & 23369648839 & 10067395646 & 26213362722 \\ 0 & 9 & 0 & 6 & 6 & 3 & 4 & 0 & 8 & 3 \\ 0 & 0 & 3 & 0 & 0 & 0 & 2 & 1 & 0 & 2 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 382 factors with signed exponents of absolute value up to 6194921944653; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (4 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1832)y^3 + (22s - 19966)y^2 + (-93s + 666465)y + (-360s - 1976130)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 3661y^8 - 32580y^7 + 20030890y^6 - 604930996y^5 + \\ & 8188001885y^4 - 70810902186y^3 + 413252218233y^2 - 1612277656290 \\ & *y + 5883520122900 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right)s, \quad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5239772081804 = 2^2 \cdot 13 \cdot 5039 \cdot 19996993$:

$$\begin{pmatrix} 2619886040902 & 1904951301564 & 1376921369383 & 1881009665917 & 827239693275 & 970391940115 & 1925408882982 & 962382564726 & 720046357498 & 551001009085 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 381 factors with signed exponents of absolute value up to 5997408430356; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (1 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1832)y^3 + (22s - 19966)y^2 + (-93s + 666465)y + (-360s - 1976130)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 3661y^8 - 32580y^7 + 20030890y^6 - 604930996y^5 + \\ & 8188001885y^4 - 70810902186y^3 + 413252218233y^2 - 1612277656290 \\ & *y + 5883520122900 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + \left(\frac{10043}{2626056128537293}\right)s, \quad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $31211851870487 = 31211851870487$:

$$\begin{pmatrix} 31211851870487 & 7586404893306 & 26914376810131 & 18061524010582 & 22716375521994 & 8315799643443 & 18488745765277 & 24329618889961 & 18022785923427 & 17885460220108 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 376 factors with signed exponents of absolute value up to 1884749538298; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.